TU Series Control Protocol V3.0.0

1. Change History

Date	Version	Description	Modified By	Approved By
2020-11-24	V1.0.0	Initial version	Bai Chuanchuan	
2020-12-16	V1.0.1	Modified the third-generation protocol for adjusting brightness mode.	Bai Chuanchuan	
2023-07-14	V1.1.0	Added several APIs for retrieving and setting third-generation protocols. Updated or removed some of the existing APIs.	Wang Yang	
2023-12-27	V1.2.0	Added section 2.3 field description Added section 3.5.11 program playback control Updated section 3.5.4 power off	Wen Yunlong	
2024-05-08	V2.0.1	Updated the source switching, program playback control command applicable products.	Zhang Xiaoyu	
2024-05-09	V2.0.2	Section 3.1.1 Updated link status Section 3.3.2 Updated HDMI status Section 3.3.4 Updated current display source Section 3.5.5 Updated unmute system	Zhang Xiaoyu	

		Section 3.5.9 Updated Wi-Fi station Section 3.5.11 Updated bluetooth on/off command		
2024-08-01	V2.0.5	Added HDMI3 in section 4.3.1 Current Display Source. Added HDMI3 in section 4.3.2 HDMI Status. Added HDMI3 in section 4.5.4 Switch to a Specified HDMI Source	Zhang Xiaoyu	-
2024-10-24	V3.0.0	Added the following in TU40 Pro V1.5.2： 1. Turn on/off the screen 2. Switch the effect mode Corrected the errors in this document.	Luo Chen Zong Haomai	

2. Overview

This document introduces functions of the control protocol for the TU series and the brief use of these functions.

This document also introduces how to use a debugging tool for Windows to perform common function debugging and testing. If collaborative development is required, please reach out to the appropriate developers for assistance.

3. Protocol Description

1. Request Packet

55H	AAH
ACK	Labels
Source Address	Destination Address

Device Type	Port Address
Board Address[7:0]	Board Address[15:8]
Code	Reserved
Register Unit Address[7:0]	Register Unit Address[15:8]
Register Unit Address[23:16]	Register Unit Address[31:24]
Valid Data Length[7:0]	Valid Data Length[15:8]
Write Data 0	
Write Data 1	
....	
Write Data N	
Checkout[7:0]	Checkout[15:8]

Notes:

- Each cell represents one byte.
- Currently, the checkout field is not validated.

2. Response Packet

AAH	55H
ACK	Labels
Source Address	Destination Address
Device Type	Port Address
Board Address[7:0]	Board Address[15:8]
Code	Reserved
Register Unit Address[7:0]	Register Unit Address[15:8]
Register Unit Address[23:16]	Register Unit Address[31:24]
Valid Data Length[7:0]	Valid Data Length[15:8]
Write Data 0	
Write Data 1	
....	
Write Data N	
Checkout[7:0]	Checkout[15:8]

Notes:

Each cell represents one byte.

3. Field Description

Field	Length (byte)	Meaning
Package Header	2	55AAH indicates a request packet. AA55H indicates a response packet
ACK	1	Response status code 0x00: ok 0x01: error 0x02: Request packet validation error 0x03: Response packet validation error 0x04: Unsupported command
Labels	1	Command serial number, default to 0x0
Source Address	1	Command source address Terminal: 0xfc Remote: 0x00
Destination Address	1	Command destination address 0xfc: Terminal device 0x00: Remote device
Device Type	1	The device type corresponds to the operational functions described in the document. This field has high maintenance priority, and any addition of new operational functions requires an update here. 0x00 3.1 Sending Card 06x01 3.2 Receiving Card 0x02 Multifunction Card 0x07 3.4 Remote Control Functions 00 0x08 3.5 Android Card Control Functions
Port Address	1	Ethernet port address
Board Address	2	The address of the receiving card connected to the Ethernet port.
Code	1	Transaction code

		0x00 Read 0x01 Write
Reserved	1	Reserved field, default to 0x00
Register Unit Address	4	Operational function tags correspond to the sub-functions under each operational function (Device Type), with maintenance permissions assigned to each specific operational function. For example, in section 4.3.1 "USB Drive Status," the device type is 0x06, and the register unit address is 0x01.
Valid Data Length	2	The length of the payload valid data. Not present in read request packets or write response packets.
Data	NA	Valid data, the data length should correspond to the valid data length.
Checkout	2	Checksum. The checksum is obtained by summing all the data bytes (excluding the package header) and adding the constant 0x5555

4. Supported Functions and Their Descriptions

1. Sending Card Functions

Link Status

- Ethernet ports 1-8

  Read: 55AA0001FC000000000000002A00000001007D56

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 00 7D 56 no link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 01 7E 56 port 1 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 02 7F 56 port 2 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 03 80 56 ports 1 and 2 link

Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 04 81 56 port 3 link

Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 08 85 56 port 4 link

Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 10 8D 56 port 5 link

Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 20 9D 56 port 6 link

Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 40 9D 56 port 7 link

Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 01 00 80 9D 56 port 8 link

- Ethernet ports 9-16

  Read: 55AA0001FC000000000000002D00000001008256

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 01 00 7E 56 port 9 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 02 00 7F 56 port 10 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 04 00 80 56 port 11 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 08 00 81 56 port 12 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 10 00 85 56 port 13 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 20 00 8D 56 port 14 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 40 00 9D 56 port 15 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 2A 00 00 00 02 00 80 00 9D 56 port 16 link

- Ethernet ports 17-20

  Read: 55AA0001FC000000000000003F00000001008256

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 3F 00 00 00 01 00 00 7D 56 no link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 3F 00 00 00 01 00 01 7E 56 port 17 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 3F 00 00 00 01 00 02 7F 56 port 18 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 3F 00 00 00 01 00 03 80 56 ports 17 and 18 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 3F 00 00 00 01 00 04 81 56 port 19 link

  Response: AA 55 00 01 00 FC 00 00 00 00 00 00 3F 00 00 00 01 00 08 85 56 port 20 link

2. Receiving Card Functions

1. Brightness

1. Set the receiving card brightness to 0x32

   Write: 55AA0001FC0001FFFFFF0100010000020100328759

   Response: AA 55 00 01 00 FC 01 FF FF FF 01 00 01 00 00 02 00 00 54 59
2. Read the brightness value of the first receiving card connected to Ethernet port 1 of the sending card:

   Read: 55 AA 00 01 FC 00 01 00 00 00 00 00 01 00 00 02 01 00 57 56

   Response: AA 55 00 01 00 FC 01 00 00 00 00 00 01 00 00 02 01 00 32 89 56

2. Voltage

Read the voltage of the first receiving card connected to Ethernet port 1 of the sending card

Read: 55AA0001FC000100000000000300000A01006156

Response: AA 55 00 01 00 FC 01 00 00 00 00 00 03 00 00 0A 01 00 A9 0A 57

Note: Taking the lower 7 bits of A9, it is 41. The unit is 0.1 V. Therefore, the actual value is 4.1 V.

3. Temperature

Read the temperature of the first receiving card connected to Ethernet port 1 of the sending card

Read: 55AA0001FC000100000000000100000A01005F56

Response: AA 55 00 01 00 FC 01 00 00 00 00 00 01 00 00 0A 01 00 48 A7 56

Note: Converting 48 to a decimal, it is 72. The unit is 0.5°C. Therefore, the actual value is 36°C.

3. Monitoring Functions at the Android Card End

1. USB Drive Status

Read: 55AA0001FC00060000000000000100000001005A56

Response: AA55000100FC0600000000000001000000010001 5B56 USB drive inserted

AA55000100FC0600000000000001000000010000 5A56 USB drive not inserted

Note: 1: The USB drive is inserted; 0: The USB drive is not inserted.

2. HDMI Status

Read: 55AA0001FC00060000000000000200000004005E56

Response: AA55000100FC06000000000002000000040000000000 5E56 No HDMI connected

AA55000100FC060000000000020000000400010000005F56 HDMI1 connected

AA55000100FC060000000000020000000400000100005F56 HDMI2 connected

AA55000100FC060000000000020000000400000001005F56 Android connected

AA55000100FC060000000000020000000400000000015F56 HDMI3 connected

Note: For the four fields highlighted in red, the first field indicates whether HDMI1 is connected, the second field indicates whether HDMI2 is connected, the third field indicates whether Android is connected, and the fourth field indicates whether HDMI3 is connected. Multiple. connections can be active simultaneously.

01 represents connected, 00 represents not connected

3. Screen Mirroring Status

Read: 55AA0001FC000600000000000003000000001005C56

Response: AA55000100FC0600000000000300000001000005C56 (No screen mirroring)

AA55000100FC0600000000000300000001000015D56 (1 screen mirroring)

Note: The data in red indicates the quantity of accessed screen mirroring.

4. Current Display Source

Read: 55AA0001FC000600000000000004000000001005D56

Response: AA55000100FC0600000000000400000001000005D56 ANDROID source

AA55000100FC0600000000000400000001000015E56 HDMI1 source

AA55000100FC0600000000000400000001000025F56 HDMI2 source

AA55000100FC0600000000000400000001000035F56 HDMI3 source

Note:

0 represents the Android source

1 represents the HDMI1 source

2 represents the HDMI2 source

3 represents the HDMI3 source

5. System Volume

Read: 55AA0001FC000600000000000005000000002005F56

Response: AA55000100FC06000000000005000000020003299156 (Not muted, volume at 50)

AA55000100FC06000000000005000000020013299256 (Muted, volume at 50)

Note: For the byte array in red, the data that has index 0 indicates whether it is mute (1: mute; 0: not mute). The data that has index 1 indicates the volume. The data that is read is hexadecimal. Please convert it to a decimal.

6. Wi-Fi Status

Read: 55AA0001FC0006000000000000600000003006156

Response: AA55000100FC060000000000006000000030002013A9E56 (Wi-Fi turned on and connected. Signal strength at -58)

AA55000100FC06000000000000600000003000100006256 (Wi-Fi turned off)

Note: For the byte array in red, the value at index 0 indicates the Wi-Fi status (0: unknown; 1: turned off; 2: turned on; 3: turning off; 4: turning on).

The value at index 1 indicates Wi-Fi connection (1: connected; 0: not connected).

The value at index 2 indicates Wi-Fi signal strength

(This value is the absolute value of the actual value. The strength is expressed as a negative number).

7. Wired Network

Read: 55AA0001FC0006000000000000700000001006056

Response: AA55000100FC0600000000000070000000100006056 (Wired network not connected)

AA55000100FC0600000000000070000000100016156 (Wired network connected)

Note: 0: not connected; 1: connected.

8. Hotspot

Read: 55AA0001FC0006000000000000800000003006156

Response: AA55000100FC06000000000000800000001000100000XXXX (Hotspot enabled|Frequency band|Channel)

AA55000100FC0600000000000080000000100FF0000XXXX (An exception occurred while accessing the system service.)

Note: 0: Off; 1: On.

9. Bluetooth

Read: 55AA0001FC0006000000000000900000002006356

Response: AA55000100FC06000000000000900000002000020166656 (Bluetooth turned on but not discoverable. Can be connected by previously paired devices)

AA55000100FC0600000000000090000000200000006356 (Bluetooth turned off, not discoverable, and cannot discover other devices)

AA55000100FC0600000000000090000000200020226756 (Bluetooth turned on, both discoverable and can be connected to other devices)

Note: For the byte array in red, the value at index 0 indicates the Bluetooth status (0: turned off; 1: turning on; 2: turned on; 3: turning off). The value at index 1 indicates the Bluetooth discoverability (0: Not discoverable and cannot be connected; 1: Not discoverable, but can be connected by previously paired devices; 2: Discoverable and can be connected by other devices.)

10. Screen Mirroring Activation Status

Read: 55AA0001FC000600000000000A00000001006356

Response: AA55000100FC0600000000000A0000000100006456 (Not activated)

Response: AA55000100FC0600000000000A0000000100016556 (Trial activated)

Response: AA55000100FC0600000000000A0000000100026656 (Commercially activated)

Response: AA55000100FC0600000000000A0000000100FF (Failed to retrieve screen mirroring service)

Note: FF: Failed to retrieve screen mirroring service; 00: Not activated; 01: Trial activated; 02: Commercially activated.

11. Retrieve System Mute Status

Read: 55AA0001FC000600000000000B00000001006456

Response: AA55000100FC0600000000000B0000000100016556 (Muted)

Response: AA55000100FC0600000000000B0000000100006456 (Not muted)

Note: FF: Failed to retrieve Audio service; 00: Muted; 01: Not muted.

12. Retrieve Whether the Wired Network is Using DHCP

Read: 55AA0001FC000600000000000C00000001006556

Response: AA55000100FC0600000000000C0000000100016656 (DHCP)

Response: AA55000100FC0600000000000C0000000100006556 (Static)

13. Retrieve System Status

Read: 55AA0001FC000600000000000D00000001006656

Response: AA55000100FC0600000000000D0000000100FF6557 (Power service failed)

Response: AA55000100FC0600000000000D0000000100016756 (Normal display)

Response: AA55000100FC0600000000000D0000000100006656 (Standby)

Note: FF: Failed to retrieve Power service; 00: Standby; 01: Normal display.

4. Remote Control Functions

1. Home

Write: 55AA0001FC00070000000100010000000100005C56

Response: AA55000100FC070000000100010000000000005B56 Execution successful

AA55010100FC070000000100010000000000005C56 Execution failed

2. Back

Write: 55AA0001FC00070000000100020000000100005D56

Response: AA55000100FC070000000100020000000000005C56 Execution successful

AA55010100FC070000000100020000000000005D56 Execution failed

3. Menu

Write: 55AA0001FC00070000000100030000000100005E56

Response: AA55000100FC070000000100030000000000005D56 Execution successful

AA55010100FC070000000100030000000000005E56 Execution failed

4. Power Button

Write: 55AA0001FC00070000000100040000000100005F56

Response: AA55000100FC070000000100040000000000005E56 Execution successful

AA55010100FC070000000100040000000000005F56 Execution failed

5. Switch Signal Source

Write: 55AA0001FC000700000001000500000001000006056

Response: AA55000100FC070000000100050000000000005F56 Execution successful

AA55010100FC0700000001000500000000000006056 Execution failed

6. Volume Up

Write: 55AA0001FC000700000001000600000001000006156

Response: AA55000100FC0700000001000600000000000006056 Execution successful

AA55010100FC0700000001000600000000000006156 Execution failed

7. Volume Down

Write: 55AA0001FC000700000001000700000001000006256

Response: AA55000100FC0700000001000700000000000006156 Execution successful

AA55010100FC0700000001000700000000000006256 Execution failed

8. Up

Write: 55AA0001FC00070000000100080000000100006356

Response: AA55000100FC0700000001000800000000006256 Execution successful

AA55010100FC0700000001000800000000006356 Execution failed

9. Down

Write: 55AA0001FC0007000000010009000000010000 6456

Response: AA55000100FC0700000001000900000000006356 Execution successful

AA55010100FC0700000001000900000000006456 Execution failed

10. Left

Write: 55AA0001FC000700000001000A000000010000 6556

Response: AA55000100FC070000000 1000A00000000006456 Execution successful

AA55010100FC0700000001000A00000000006556 Execution failed

11. Right

Write: 55AA0001FC000700000001000B000000010000 6656

Response: AA55000100FC0700000001000B00000000006556 Execution successful

AA55 0101 00FC0700000001000B00000000006656 Execution failed

12. Confirm

Write: 55AA0001FC000700000001000C000000010000 6756

Response: AA55000100FC0700000001000C00000000006656 Execution successful

AA55010100FC0700000001000C00000000006756 Execution failed

5. Android Card Control Functions

1. Standby

Write: 55AA0001FC00080000000100010000000100005D56

Response: AA55000100FC0800000001000100000000005C56 Execution successful

AA55010100FC0800000001000100000000005D56 Execution failed

2. Wake Up from Standby

Write: 55AA0001FC00080000000100020000000100005E56

Response: AA55000100FC0800000001000200000000005D56 Execution successful

AA55010100FC0800000001000200000000005E56 Execution failed

3. Power Off

Write: 55AA0001FC000800000001000300000000100005F56

Response: AA55000100FC080000000100030000000000005C56 Execution successful

AA55010100FC080000000100030000000000005D56 Execution failed

4. Switch to a Specified HDMI Source

1. HDMI1

   Write: 55AA0001FC0008000000010004000000001000016156

   Response: AA55000100FC0800000001000400000000005F56 Execution successful

   AA55010100FC0800000001000400000000006056 Execution failed

2. HDMI2

   Write: 55AA0001FC0008000000010004000000001000026256

   Response: AA55000100FC0800000001000400000000005F56 Execution successful

   AA55010100FC0800000001000400000000006056 Execution failed

3. ANDROID

   Write: 55AA0001FC0008000000010004000000001000036356

   Response: AA55000100FC0800000001000400000000005F56 Execution successful

   AA55010100FC0800000001000400000000006056 Execution failed

4. HDMI3

   Write: 55AA0001FC0008000000010004000000001000046356

   Response: AA55000100FC0800000001000400000000005F56 Execution successful

   AA55010100FC0800000001000400000000006056 Execution failed

5. Mute/Unmute the System

1. Mute

   Write: 55AA0001FC00080000000100050000000100015E56

   Response: AA55000100FC0800000001000500000000005D56 Execution successful

   AA55010100FC0800000001000500000000005E56 Execution failed

2. Unmute

   Write: 55AA0001FC00080000000100050000000100005E56

   Response: AA55000100FC0800000001000500000000005F56 Execution successful

   AA55010100FC0800000001000500000000006056 Execution failed

6. Set System Volume

Write: 55AA0001FC00080000000100060000000100335E56

Response: AA55000100FC0800000001000600000000005F56 Execution successful

AA55010100FC0800000001000600000000006056 Execution failed

Note:

Volume adjustable range: 0 to 100 or 0x00 to 0x64. If the value exceeds 100, it will be set as 100.

7. Turn On/Off the Relay

Write: 55AA0001FC000800000001000700000002000015E56

Response: AA55000100FC080000000100070000000005F56 Execution successful

AA55010100FC080000000100070000000006056 Execution failed

Note:

For the 2 bytes used to turn on/off the relay, the first byte indicates the relay ID (default to 0) and the second byte indicates on/off (0: off; 1: on).

8. Eco Mode Setting

(Currently not supported currently due to implementation constraints)

Write: 55AA0001FC0008000000010008000000010005E56

Response: AA55000100FC080000000100080000000005F56 Execution successful

AA55010100FC080000000100080000000006056 Execution failed

9. Turn On/Off Wi-Fi Station

Write: 55AA0001FC0008000000010009000000100015E56

Response: AA55000100FC080000000100090000000005F56 Execution successful

AA55010100FC080000000100090000000006056 Execution failed

Note: 01: On; 00: Off

10. Turn On/Off Wi-Fi AP

Write: 55AA0001FC000800000001000A000000100015E56

Response: AA55000100FC08000000001000A00000000005F56 Execution successful

AA55010100FC08000000001000A00000000006056 Execution failed

Note: 01: On; 00: Off

11. Turn On/Off Bluetooth

- Bluetooth On

  Write: 55AA0001FC000800000001000B000000100015E56

  Response: AA55000100FC08000000001000B00000000005F56 Execution successful

  AA55010100FC08000000001000B00000000006056 Execution failed

- Bluetooth Off

  Write: 55AA0001FC000800000001000B000000100005E56

Response: AA55000100FC0800000001000B00000000005F56 Execution successful

AA55010100FC0800000001000B00000000006056 Execution failed

12. Program Playback Control

To perform program playback control operations, the program list must be initialized first. After initialization, the device will return the total number of programs to the remote end. At this point, the remote end can begin controlling program playback, such as starting, pausing, resuming, etc. Program operations are performed using program indices, starting from 1 and not exceeding the total number of programs.

1. Initialize Program List

   The maximum number of programs is 0xFF, represented by one byte. If the total number of programs exceeds 0xFF, only 0xFF will be returned.

   Read: 55AA0001FC000800000000000C00000001005E56

   FF represents the number of programs (for example purposes only, it is a real value)

   Response: AA55000100FC08000000000000C000000000100FF5F56 List initialization successful

   AA55010100FC0800000000000C00000000006056 Failed

2. Play a Specified Program

   0x0201 in the data area corresponds to playing the first program, while 0x0202 indicates playing the second program.

   Write: 55AA0001FC00080000000100 0C00000002000201 5E56

   Response: AA55000100FC0800000001000C00000000005F56 Execution successful

   AA55010100FC0800000001000C00000000006056 Execution failed

3. Pause the Program Being Played

   Write: 55AA0001FC000800000001000C00000001000035E56

   Response: AA55000100FC0800000001000C00000000005F56 Execution successful

   AA55010100FC0800000001000C00000000006056 Execution failed

4. Resume Playback

   Write: 55AA0001FC000800000001000C00000001000045E56

   Response: AA55000100FC0800000001000C00000000005F56 Execution successful

   AA55010100FC0800000001000C00000000006056 Execution failed

5. Exit the Playing/Paused Program

   Write: 55AA0001FC000800000001000C00000001000055E56

   Response: AA55000100FC0800000001000C00000000005F56 Execution successful

   AA55010100FC0800000001000C00000000006056 Execution failed

6. Switch to the Previous Program on the List

   Write: 55AA0001FC000800000001000C00000000100065E56

   Response: AA55000100FC0800000001000C00000000005F56 Execution successful

   AA55010100FC0800000001000C00000000006056 Execution failed

7. Switch to the Next Program on the List

   Write: 55AA0001FC000800000001000C00000000100075E56

   Response: AA55000100FC0800000001000C00000000005F56 Execution successful

   AA55010100FC0800000001000C00000000006056 Execution failed

13. Turn On/Off the Screen

Write: 55AA0001FC000800000001000D00000000100015E56

Response: AA55000100FC0800000001000D00000000006856 Execution successful

AA55010100FC0800000001000D00000000006956 Execution failed

Note: 01: On; 00: Off

14. Switch the Effect Mode

Write: 55AA0001FC000800000001000E00000000100015E56

Response: AA55000100FC0800000001000E00000000006956 Execution successful

AA55010100FC0800000001000E00000000006A56 Execution failed

Note:

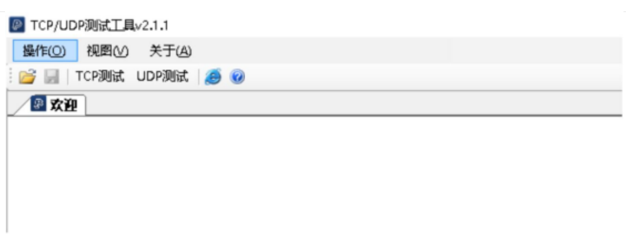
Standard: 0; Meeting: 1; Vivid: 2; Skin: 3

5. Instructions for the Windows Debugging Tool

Step1: Unzip the package in the attachment. (tcpudp.rar)

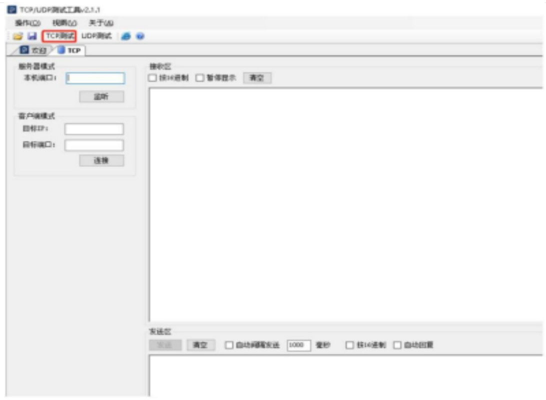
Step2: Double click tcpudp_2.1.1.exe in the folder

Step3: If it runs normally, the following interface will appear.

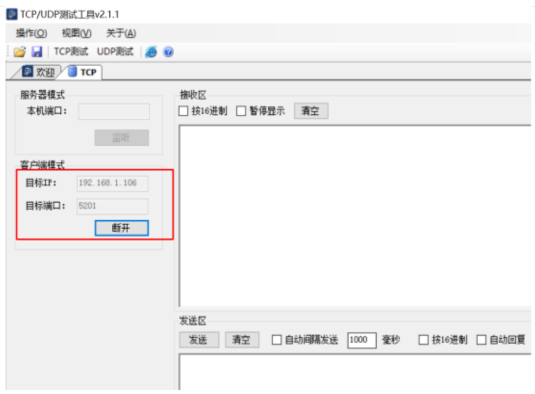


Step4: Connect to TCP.

1. Click **TCP Test** to access the following interface.



2. In client mode, enter the target IP address (the IP address of the meeting board) and the target port address (default to 5201), then click **Connect**. Refer to the figure below:

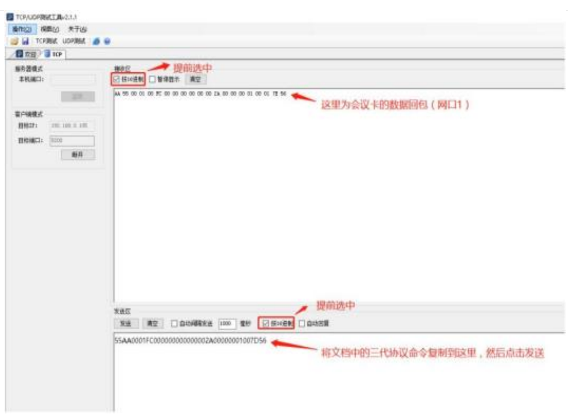


Note: If the connection is normal, the **Connect** button will become a **Disconnect** button, as shown in the figure above.

3. Perform command testing.

After the above steps are done, the TCP connection is established, and the relevant functions of control can be tested.

Take reading the link status of the sending card as an example:



Note:

This TCP/UDP tool is a third-party tool downloaded from the Internet, and may be not stable (exiting unexpectedly which happens occasionally). In case of unexpected exit, simply restart the TCP/UDP tool and reconnect again.